

Teaching logical statements at the undergraduate level

Why one common definition gets in the way

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A recurring classroom experience

I have tutored undergraduates working through introductory logic—from Rosen's *Discrete Mathematics* and similar texts.

A pattern I kept noticing:

- Students were comfortable reading $x + 1 = 2$ as an English sentence.
- They struggled to treat it as a *formal object* to be manipulated—negated, quantified, combined.
- The formula always had to *mean something* before anything could be done with it.

Where was this coming from?

The Day-1 definition

From Rosen's *Discrete Mathematics and Its Applications*:

“A proposition is a declarative sentence that is either true or false, but not both.”

Rosen then immediately says:

- $1 + 1 = 2$ ✓ a proposition
- $x + 1 = 2$ × *not* a proposition—neither true nor false

Open expressions are thus marked, on Day 1, as logically *illegitimate*.

What this definition quietly does

The definition ties “being a statement” to having a definite truth value.

But this conflates two different things:

Form	Meaning
The shape of the sentence; what we can write down and manipulate.	Whether the sentence is true or false in some interpretation.

A sentence can have a *perfectly good form* for logical manipulation even if its truth value is not yet determined.

What students do as a result

Because $x > 0$ “isn’t a statement,” students feel they must *repair* it before using it.

The most common repair: attach an implicit universal quantifier.

$$x > 0 \quad \longrightarrow \quad \forall x \in \mathbb{R}, x > 0$$

This seems harmless—until it isn’t.

A recent summer school for first-year undergraduates

A faculty member presented these as equivalent ways of writing the same statement:

$$\forall x \in \mathbb{R}, x^2 \geq 2x \quad \text{is the same as} \quad x \in \mathbb{R} \implies x^2 \geq 2x$$

A student immediately asked: *where did the $\forall x$ go?*

The answer given: *“it’s there implicitly.”*

Why “it’s there implicitly” is not a good answer

The two forms *are* related, but not as notational variants:

$$\forall x \in \mathbb{R}, P(x) \quad \text{corresponds to} \quad \forall x, (x \in \mathbb{R} \implies P(x))$$

The quantifier is not implicit inside the implication—it must bind *outside* it.

If students (and instructors) believe that a free variable always carries an invisible $\forall$, they will:

- misread what “let $x \in \mathbb{R}$ ” declares in a proof,
- conflate universally quantified statements with implications,
- and—further down the road—struggle seriously with mathematical logic.

These are not exotic errors. They are a natural consequence of the Day-1 definition.

A concrete illustration

At a foundations workshop, students were asked:

What is the negation of “The square of any integer is an integer”?

One of the options offered was:

If $n^2 \notin \mathbb{Z}$, then $n \notin \mathbb{Z}$.

Two instructors disagreed on whether this option *was* the original statement or its contrapositive.

A concrete illustration (cont'd)

The original statement is:

$$\forall n \in \mathbb{Z}, n^2 \in \mathbb{Z}$$

a universally quantified statement, *not* an implication.

So the option offered is neither the statement, nor its negation, nor its contrapositive.

The disagreement only arises because both instructors had silently rewritten the original as

$$\text{If } n \in \mathbb{Z}, \text{ then } n^2 \in \mathbb{Z},$$

with $\forall n$ left implicit — exactly the ghost-quantifier habit at work.

This is not a beginner's error. It trips up experienced educators too.

A small fix

Instead of defining “statement,” show two columns of examples.

Sentences we will reason about	Sentences we will NOT reason about
0 is a natural number.	What a beautiful proof! (<i>exclamation</i>)
The sun rises in the west.	Is π irrational? (<i>question</i>)
$x > 0$	Please close the door. (<i>command/request</i>)
$\forall n \in \mathbb{Z}, n^2 \geq 0$	Maybe it will rain. (<i>possibility</i>)
He is a rich man.	

We will negate, combine, quantify, and reason with everything in the left column—using the same tools throughout.

What this changes

- **Open expressions are first-class.**

$x > 0$ appears in the left column from Day 1. No repair needed before reasoning with it.

- **Quantifiers do real work.**

Adding $\forall x \in \mathbb{R}$ is a genuine act with a genuine effect—not a formality to legitimise a formula.

- **The cost is almost nothing.**

Two columns of examples take the same time as one definition. Nothing downstream needs to change.

Why the size of the fix is misleading

The change is small. Its consequences are not.

A student who treats open expressions as legitimate objects will:

- read $\forall x \in \mathbb{R}, P(x)$ and $\exists x \in \mathbb{R}, P(x)$ as genuinely different claims about $P(x)$,
- write proofs in which “let $x \in A$ ” and “for all $x \in A$ ” play distinct roles,
- and approach mathematical logic—should they ever meet it—without unnecessary conceptual baggage.

Serious errors in logical reasoning can sometimes be corrected simply by not saying the wrong thing on Day 1.

Summary

Current approach	Proposed approach
Define: a statement is a sentence with an unambiguous truth value.	Show two columns: sentences we will reason about, and sentences we won't.
Open expressions feel illegitimate until quantified.	Open expressions are in the left column from the start.
Implicit quantifiers appear to patch the gap.	Quantifiers do explicit, meaningful work.

*A genuinely small change.
Its effect on how students work with quantifiers and proofs
is, in my experience, not small at all.*